

Advanced Econometrics & Dynamic Theory

Problem Set 3: Causal Inference & Dynamic Optimization

Instructor: Prof. Aris Thorne | Department of Economics

Course Code: ECON 4120 / 6120
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 Submission Due: March 28, 2026 (23:59 EST)
 Target Audience: Senior Undergrad / Graduate

Student Name: _____

Student ID / Matriculation: _____

Instructions: Show all derivation steps for analytical questions. Attach executable R scripts for empirical sections.

Score / Total: / 100

1 Part I: Intertemporal Choice & Liquidity Constraints

Problem 1: Household Consumption Smoothing under Borrowing Limits (25 Points)

Consider a two-period consumption problem. A household receives endowment $y_1 > 0$ in Period 1 and $y_2 > 0$ in Period 2, maximizing CRRA utility $U(c_1, c_2) = \frac{c_1^{1-\gamma} - 1}{1-\gamma} + \beta \left(\frac{c_2^{1-\gamma} - 1}{1-\gamma} \right)$ with $\gamma > 0, \gamma \neq 1, \beta \in (0, 1)$, subject to interest rate $r > 0$ and borrowing limit $c_1 \leq y_1$.

(a) Formulate the constrained optimization problem using the Lagrangian method and derive the modified Euler equation.

Solution: Writing the Lagrangian with borrowing constraint multiplier $\lambda \geq 0$:

$$\mathcal{L} = \frac{c_1^{1-\gamma} - 1}{1-\gamma} + \beta \left(\frac{c_2^{1-\gamma} - 1}{1-\gamma} \right) + \mu \left[y_1 + \frac{y_2}{1+r} - c_1 - \frac{c_2}{1+r} \right] + \lambda (y_1 - c_1)$$

First-order conditions $\frac{\partial \mathcal{L}}{\partial c_1} = c_1^{-\gamma} - \mu - \lambda = 0$ and $\frac{\partial \mathcal{L}}{\partial c_2} = \beta c_2^{-\gamma} - \frac{\mu}{1+r} = 0$ yield the modified Euler Equation:

$$c_1^{-\gamma} = \beta(1+r)c_2^{-\gamma} + \lambda \quad (1)$$

Complementary slackness requires $\lambda \geq 0, (y_1 - c_1) \geq 0$, and $\lambda(y_1 - c_1) = 0$. When the borrowing limit binds ($\lambda > 0$), $c_1^* = y_1, c_2^* = y_2$, and marginal utility satisfies $c_1^{-\gamma} > \beta(1+r)c_2^{-\gamma}$.

(b) Graphically illustrate optimal allocation (c_1^*, c_2^*) under (i) Unconstrained regime ($\lambda = 0$) and (ii) Binding constraint ($\lambda > 0$).

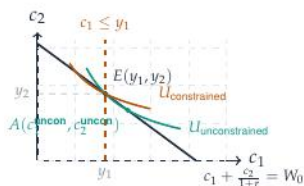


Figure 1: Intertemporal Consumption Choice under Borrowing Constraints. Low y_1 causes unconstrained point A to violate $c_1 \leq y_1$, forcing consumption to endowment point E.